

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0798

East Beloit 230 kV
May 27, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0798. The Interconnection Customer proposes a 79.0 MW solar photovoltaic facility interconnecting at the East Beloit 230 kV substation in the Illinois area. The request was received on May 17, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 363 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,707,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: EUFAULA 230/1kV Auto
Contingency: IMPERIAL SUBSTATION-EUFAULA 500kV line
Loading: 115.0% (Rating: 1,453 MVA)
DFAX: 12.4%
Pre-project loading: 114.4%
2. Facility: EUFAULA 500/1 kV Transformer
Contingency: IMPERIAL-EUFAULA 500kV L
Loading: 115.0% (Rating: 1,453 MVA)
DFAX: 12.4%
Pre-project loading: 114.4%
3. Facility: WAYNE TRACE SUBSTATION-DARLINGTON RD 100kV
Contingency: RICHLAND-DEKALB 500kV Line
Loading: 114.5% (Rating: 258 MVA)
DFAX: 6.0%
Pre-project loading: 112.7%
4. Facility: JOSO-ONANCOCK DIESEL GENERATION FACILITY 230kV
Contingency: BEAVER-EUFAULA 500kV
Loading: 113.7% (Rating: 1,316 MVA)
DFAX: 14.4%
Pre-project loading: 112.9%
5. Facility: POST ROCK-NEWPORT TIE 230kV
Contingency: TREGO-BEAVER 500kV
Loading: 113.1% (Rating: 1,358 MVA)
DFAX: 13.4%
Pre-project loading: 112.3%

6. Facility: ONANCOCK DIESEL GENERATION FACILITY-BEAVER 230kV
Contingency: BEAVER-EUFAULA 500kV
Loading: 113.0% (Rating: 1,315 MVA)
DFAX: 11.0%
Pre-project loading: 112.4%
7. Facility: POST ROCK-MYRTLE PT 230kV line
Contingency: TREGO-BEAVER 500kV
Loading: 112.6% (Rating: 1,406 MVA)
DFAX: 12.9%
Pre-project loading: 111.8%
8. Facility: LIONS RD-PINEY CRK PROJECT SUBSTATION 500kV
Contingency: EL PSO SUBSTATION-GANSO 500kV line
Loading: 112.4% (Rating: 3,166 MVA)
DFAX: 12.6%
Pre-project loading: 112.1%
9. Facility: P.T.S-MEDORA 345kV
Contingency: SUGAR HILL-ALLEGHENY ENERGY UNITS 3 4 & 5 345kV Line
Loading: 111.7% (Rating: 1,894 MVA)
DFAX: 7.6%
Pre-project loading: 111.3%
10. Facility: MUSKOGEE-DARLINGTON RD 100kV line
Contingency: RICHLAND-DEKALB 500kV
Loading: 111.5% (Rating: 231 MVA)
DFAX: 6.7%
Pre-project loading: 109.2%
11. Facility: MSON ST-NEW DEPTURE 230kV
Contingency: TREGO-BEAVER 500kV
Loading: 111.1% (Rating: 842 MVA)
DFAX: 7.9%
Pre-project loading: 110.4%
12. Facility: BURWELL 500/345kV Transformer Bank
Contingency: ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV Transformer
Loading: 111.0% (Rating: 1,191 MVA)
DFAX: 6.1%
Pre-project loading: 110.6%
13. Facility: ADAMS-PICKFORD 230kV
Contingency: BEAVER-TREGO 500kV
Loading: 110.8% (Rating: 855 MVA)
DFAX: 8.1%
Pre-project loading: 110.1%
14. Facility: WEEDPATCH-CANAL RD 230kV
Contingency: TREGO-BEAVER 500kV
Loading: 109.5% (Rating: 966 MVA)
DFAX: 7.7%
Pre-project loading: 108.9%
15. Facility: HANDSOME LAKE-PINEY CRK PROJECT 500kV
Contingency: LIONS RD-PINEY CRK PROJECT 500kV line
Loading: 109.2% (Rating: 2,747 MVA)
DFAX: 12.1%
Pre-project loading: 108.8%
16. Facility: HLLTOP SUBSTATION 500-345 kV XFMR
Contingency: HILLTOP-TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500kV Line

- Loading: 109.1% (Rating: 2,218 MVA)
DFAX: 7.0%
Pre-project loading: 108.9%
17. Facility: ONANCOCK DIESEL GENERATION FACILITY-ADAMS 230kV Line
Contingency: TREGO-BEAVER 500kV Line
Loading: 108.3% (Rating: 1,686 MVA)
DFAX: 16.1%
Pre-project loading: 107.5%
18. Facility: CARBON CTR 500-345 kV Autotransformer
Contingency: BLACKROCK SUBSTATION-SHAWVILLE STA 345kV
Loading: 108.0% (Rating: 1,716 MVA)
DFAX: 6.4%
Pre-project loading: 107.7%
19. Facility: TALENENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV XFMR
Contingency: WESTWOOD GENERATION LLC-CORONET 500kV line
Loading: 107.9% (Rating: 1,584 MVA)
DFAX: 5.9%
Pre-project loading: 107.6%
20. Facility: STEWART MTN DAM-RITTER PARK 345kV
Contingency: MEDORA-P.T.S 345kV line
Loading: 107.0% (Rating: 1,760 MVA)
DFAX: 7.0%
Pre-project loading: 106.7%
21. Facility: CONESVILLE STATION-DRESDEN 500kV
Contingency: PT ARENA-DRESDEN 500kV
Loading: 106.8% (Rating: 3,582 MVA)
DFAX: 5.9%
Pre-project loading: 106.7%
22. Facility: ORCHARD RD-SHAWVILLE STA SUBSTATION 500kV
Contingency: HILLTOP-PINEY CRK PROJECT 500kV Line
Loading: 106.7% (Rating: 2,560 MVA)
DFAX: 8.8%
Pre-project loading: 106.4%
23. Facility: P.T.S-BURWELL 345kV
Contingency: ALLEGHENY ENERGY UNITS 3 4 & 5 SUBSTATION 345/1kV Autotransformer
Loading: 106.6% (Rating: 1,721 MVA)
DFAX: 7.1%
Pre-project loading: 106.2%
24. Facility: HAMPTON RD-MEDORA 345kV Line
Contingency: FISHKILL PLAINS-SIMPSON STUART TAP 345kV
Loading: 106.4% (Rating: 1,838 MVA)
DFAX: 6.9%
Pre-project loading: 106.1%
25. Facility: PICKFORD-HERALD-TWIN CITIES SUBSTATION 230kV
Contingency: TREGO-BEAVER 500kV
Loading: 106.2% (Rating: 1,138 MVA)
DFAX: 9.8%
Pre-project loading: 105.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EUFAULA 230/1kV TX	Replace transformer with higher-rated unit	\$9,579,000	1% (\$105,000)
EUFAULA 500/1 kV Transformer	Replace transformer with higher-rated unit	\$28,636,000	1% (\$315,000)
WAYNETRACE-DARLINGTON RD 100kV	Rebuild line with higher-capacity conductor	\$2,655,000	1% (\$29,000)
JOSO-ONANCOCK DIESEL GENERATION FACILITY 230kV	Rebuild line with higher-capacity conductor	\$33,807,000	1% (\$372,000)
POST ROCK-NEWPORT TIE 230kV	Rebuild line with higher-capacity conductor	\$52,068,000	1% (\$572,000)
BEAVER-ONANCOCK DIESEL GENERATION FACILITY 230kV	Rebuild line with higher-capacity conductor	\$24,683,000	1% (\$271,000)
POST ROCK-MYRTLE PT 230kV line	Rebuild line with higher-capacity conductor	\$19,711,000	1% (\$217,000)
LIONS RD-PINEY CRK PROJECT SUBSTATION 500kV	Rebuild line with higher-capacity conductor	\$139,529,000	1% (\$1,542,000)
MEDORA SUBSTATION-P.T.S 345kV	Rebuild line with higher-capacity conductor	\$30,635,000	1% (\$337,000)
MUSKOGEE-DARLINGTON RD 100kV line	Rebuild line with higher-capacity conductor	\$20,728,000	1% (\$229,000)
NEW DEPARTURE-MASON ST 230kV Line	Rebuild line with higher-capacity conductor	\$37,900,000	1% (\$417,000)
BURWELL SUBSTATION 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$28,279,000	1% (\$312,000)
ADAMS-PICKFORD 230kV	Rebuild line with higher-capacity conductor	\$19,755,000	1% (\$217,000)
CANL RD-WEEDPATCH 230kV Line	Reconductor line	\$5,379,000	1% (\$59,000)
HANDSOME LAKE-PINEY CRK PROJECT 500kV	Reconductor line	\$50,418,000	1% (\$555,000)
HILLTOP 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$25,448,000	1% (\$281,000)
ADAMS SUBSTATION-ONANCOCK DIESEL GENERATION FACILITY 230kV	Reconductor line	\$13,922,000	1% (\$153,000)
CARBON CTR 500/345 kV TX	Replace transformer with higher-rated unit	\$27,045,000	1% (\$298,000)
TALENENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$32,840,000	1% (\$364,000)
STEWART MTN DAM-RITTER PARK 345kV	Reconductor line	\$3,953,000	1% (\$44,000)
DRESDEN-CONESVILLE STATION 500kV L	Reconductor line	\$14,224,000	1% (\$151,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ORCHARD RD-SHAWVILLE STA SUBSTATION 500kV	Reconductor line	\$52,845,000	1% (\$582,000)
P.T.S-BURWELL 345kV	Reconductor line	\$4,431,000	1% (\$49,000)
HAMPTON RD-MEDORA 345kV Line	Reconductor line	\$8,869,000	1% (\$98,000)
PICKFORD-HERALD-TWIN CITIES 230kV	Reconductor line	\$12,510,000	1% (\$138,000)
	Total allocated to project		\$7,707,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.